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To cite this article: Xiwen Ouyang, Jia Zhou & Honglian Xiang (2021) Screen Mirroring is not as Easy as it Seems: A Closer Look at Older Adults' Cross-Device Experience Through Touch Gestures, International Journal of Human-Computer Interaction, 37:12, 1173-1189, DOI: [10.1080/10447318.2020.1870830](https://doi.org/10.1080/10447318.2020.1870830)

To link to this article: <https://doi.org/10.1080/10447318.2020.1870830>



Published online: 01 Feb 2021.



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Screen Mirroring is not as Easy as it Seems: A Closer Look at Older Adults' Cross-Device Experience Through Touch Gestures

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ABSTRACT

Screen mirroring might be a way to improve older adults' user experience of smart televisions (STVs) through smartphones. To examine this possibility, two experiments were conducted. Experiment I examined older adults' difficulties of screen mirroring (mirroring smartphone screens to STVs) through five common touch gestures ("Drag," "Slide," "Zoom," "Draw," and "Handwrite"), in comparison to younger adults. The results indicated that a major problem for older adults is the frequent attention switching between the STV and smartphone screens. Therefore, experiment II explored how to reduce the need of attention switching through the touch gestures ("Tap," "Slide + Tap," and "Slide + Release") and the button sizes (8, 14, and 20 mm) for different input postures. Thirty older adults participated in this experiment and their eye movements were tracked. Four major findings were derived. First, the "Zoom," "Draw," and "Handwrite" gestures in screen mirroring were difficult for older adults with a task completion rate lower than 68%. Second, the problem of frequent attention switching between the STV and smartphone was predominant for tapping tasks. Third, the "Slide + Tap" and "Slide + Release" touch gestures helped to reduce attention switching in tapping tasks more than the "Tap" for older adults, while the "Slide + Release" received the worst subjective feedback. Fourth, increasing the button size from 8 mm to 14 mm on smartphones can improve the task completion rate and the task efficiency in screen mirroring when older adults used the one-handed posture to tap.

1. Introduction

Cross-device interaction might be an alternative to improve older adults' interaction with STVs. Older adults recently can get more enjoyments by the advancement of STVs with the increased interactivity and rich resources. However, their user experiences with STVs are hindered by the traditional physical remote control because of its incomprehensible indirect interaction (McLaughlin et al., 2009), elusive button functions and complex labels (Rice & Alm, 2008), uncustomizable layout, and easily losing (Regina & Michael, 2014). Fortunately, these problems of the physical remote control can be avoided by using cross-device interactions instead. The cross-device interaction becomes an ubiquitous and convenient function due to the well equipment of Apple's Airplay or Miracast in most new mobile devices. It can be used to display contents of smartphones onto STV to enjoy the large STV screen, clear viewing experience, and direct manipulation (Sun et al., 2015), and to share the screens with others (Mcgill et al., 2014).

Among various methods of cross-device interactions, screen mirroring might be potential to bring a better STV experience to older adults. The cross-device interactions between smartphones and STVs consist of the screen mirroring (mirrored UI) and the second screen (distributed UI) according to previous reviews (Brudy et al., 2019). As for the screen mirroring, users can use it to view the activities or media contents of smartphones on the STV screen with the

direct manipulation, the simple screen relations between smartphones and STVs (the display are the same), and the rich resources of smartphones. As for the second screen, the applications of it are usually some companion ones to enhance users' tangential browsing or social viewing experience, or to act as a remote control (e.g., virtual remote applications or others, such as Apple TV Remote app) (Neate, Evans, et al., 2017). However, the second screen applications may rarely involve older people because they are too complicating for them, and the virtual remote still has the problem of indirect interaction, which cause some problems such as the inefficient focus manipulation (Ouyang & Zhou, 2018a), text input, and menu navigation (Bernhaupt et al., 2008; Carmichael, 1999; Ouyang & Zhou, 2018b). Thus, we focused on the screen mirroring in this study.

However, screen mirroring might cause new problems for older adults. The interaction with two screens simultaneously increases the difficulty of interaction and the attention distraction (Rashid et al., 2012a; Sun et al., 2015), especially for older adults. In order to allow older adults fully enjoy the advantages of screen mirroring, it is essential to find the specific critical problem of screen mirroring and then to provide the solutions.

Touch gestures are the most important and commonly used input method on smartphones; however, few researches focus on the user experience of touch gestures in the screen

mirroring. Considering the touch gestures make users able to directly manipulate the interfaces and help users to complete more complicated tasks intuitively (Sun et al., 2015), it is meaningful to investigate the performance and critical usability problems of screen mirroring under the touchscreen gestures for older adults. Thus, in this study, we focus on improving the usability of screen mirroring based on the touch gestures for older adults.

To achieve this goal, two experiments were conducted. Experiment I was conducted to investigate older adults' difficulties with touch gestures in screen mirroring. Based on the results, the critical problem identified was that older adults had to switch their attentions frequently between the smartphone and STV screen, especially when tapping on the touchscreen. Then, experiment II was conducted to verify possible solutions to this identified problem.

2. Literature review and hypotheses

2.1. The screen mirroring of cross-device interaction

Cross-device interactions become popular in recent years. According to a Google survey, 49% of users hold smartphones while watching TV (Google Inc., 2012). Similarly, 61% of Chinese users use TVs to watch popular online videos, which showed that users' cross-device interaction behaviors are becoming more common (China Internet Network Information Center, 2015). The cross-device interaction between smartphones and STVs can be divided into two types by temporal interface distribution according to the previous reviews (Brudy et al., 2019), namely, screen mirroring and second screen. With screen mirroring, the contents between smartphones and STVs are the same, while with second screen the contents are different.

The screen mirroring is a common function, which is used to mirror the smartphone screen to an STV in real time via a wired connection or a wireless transmission protocol such as DLNA, Miracast, or Airplay (Mcgill et al., 2014; Sørensen et al., 2015), or sometimes to mirror the STV screen into a smartphone in real time (Kim et al., 2013; S. Kim et al., 2013). It has two usage scenarios. Firstly, it is usually used to share users' smartphone screen with family, friends, etc., such as completing collaborative browsing tasks or consensus discussions in meetings with STVs (Mcgill et al., 2014, 2015). Secondly, users can use it to mirror the activities or media contents of the smartphone screen onto the STV for a clear view, large-screen immersive experience and directed manipulation (Sun et al., 2015), such as presenting photos and videos.

The second screen usually uses some companion applications to enhance user's viewing experience of STVs. These companion applications can be divided into three broad areas: supporting tangential browsing, supporting social viewing, and extending the remote control (Neate, Jones, et al., 2017). As for the extension of remote control, there are some virtual remote applications (e.g., Apple TV Remote app and Mi Remote app) in the consumer market, and other customizable remote which removes the interactive element from TV and transfers it to the second screen applications to control TV (Cruickshank et al., 2007). However, these extensions may increase the

difficulty of interaction and did not solve the problem caused by the indirect control of physical remote control, such as the troublesome focus manipulation (Ouyang & Zhou, 2018a) and menu navigation (Ouyang & Zhou, 2018b).

The screen mirroring has two advantages to further improve the user experience of STVs other than second screen, especially for older adults. Firstly, the interaction between the smartphone and STV of screen mirroring is easier and more intuitive for older adults because of the same contents and simple relationship between two screens than that of second screen. Secondly, with screen mirroring the contents that allows users to view are rich than that of second screen. Because screen mirroring is able to conveniently project all the contents of smartphones to STVs, while virtual remote control is usually used to manipulate the contents on the STV. Currently, the applications and resources on smartphones (e.g., Tweet, Facebook, WeChat) are more abundant than that on STVs thanks to the development of mobile internet. Therefore, it is possible that the screen mirroring might be able to bring better STV experience to older adults.

However, the screen mirroring also has usability problems. Firstly, when there is more than one screen, users have to spread their attention to different screens, and excessive attentional switches can cause significant user performance degradation (Rashid et al., 2012a; Sun et al., 2015). Secondly, older adults need to complete much excessive cognitive processing because the interaction with STVs is not direct control (Charness et al., 2004; McLaughlin et al., 2009). Thirdly, the increasing number of screens means a larger interaction scope which may result in more interference factors (Rashid et al., 2012a). So far, previous research has mostly focused on some applications and system construction of screen mirroring; however, few studies have considered the interactive problems, especially for older adults.

2.2. Touch gestures on the touchscreen

Touch gesture input is the most important and commonly used input method for smartphones, but there are few studies focusing on the touch gestures in screen mirroring. Because the interaction of screen mirroring just happens on the touchscreens of smartphones, touch gestures in screen mirroring have no difference from that on the touchscreens. Following that, the literature of this part was referred to the studies about touch gestures on smartphones only. Researches on touch gestures can be divided into two categories according to the screen size: small (e.g., smartphones) and large touchscreens (e.g., ATMs, information kiosks).

As for the touch gestures on small touchscreens, Kobayashi et al. (2011) found that older adults could complete dragging, pinching without panning, and pinching with panning tasks successfully unless the tapping buttons were smaller than 30 pixels. A possible explanation is that the ability of older adults to deal with details, move accurately (Fisk et al., 2009), as well as their time controlling (Leonardi et al., 2010) may have declined, preventing them from meeting the motion requirements of tapping. Empirically, older adults do have difficulties with the tapping gesture. They tend to press a virtual button on the touchscreen as hard as pressing a physical button,

leading to invalid taps (Chêne et al., 2016) or unintentional taps (Liu & Zhou, 2018). Additionally, once users need to tap a button in screen mirroring, they have to shift their attentions from the STV to the smartphone, which causes an unpleasant delay (Cruickshank et al., 2007; Dezfuli et al., 2014). The attentional switches between STVs and smartphones would obviously become very troublesome for users with the increasing number of taps.

As for the touch gestures on large touchscreens, a study investigated the usability of clicking, dragging, zooming, and rotating and their design features for both older adults and younger adults (Gao & Sun, 2015). The results showed that although older adults' performance of four touch gestures was worse than that for younger adults, older adults' attitudes toward these touch gestures were more positive. Accordingly, these touch gestures might have a positive effect on the user experience of screen mirroring for older adults, but its effect still needs further investigation.

2.3. Solutions to relieve attentional switches of screen mirroring

As a result of Experiment I, the frequent attentional switches in the screen mirroring under the tapping task were found problematic for older adults. Although there are some studies offered some solutions to this problem, they focused on managing user's attention of the second screen (Neate, Jones, et al., 2017), measuring users visual attention patterns (Vatavu & Mancas, 2014), surveying factors influencing visual attention switches without offering solutions (Rashid et al., 2012b). Thence, it is necessary to find the possible solutions to this problem, which could be divided into three ways.

The first way is to let users only focus on the STV screen and to use a smartphone as a second screen, such as a normal virtual remote control (Ouyang & Zhou, 2018a), the eye-free PalmRC (Ouyang & Zhou, 2018b), the system S3TV with a series of dedicated touch gestures (Sun et al., 2015), or the Touch&Screen with dedicated widgets that use simple touch gestures (Bellino, 2016). However, these solutions had some problems for older adults, for example, difficulty in remembering the gestures, complexity of operation, or inefficiency of indirect operations, which may not be suitable solutions for older adults.

The second way is to let users only focus on the smartphone screen, such as screen casting. In this way, users mainly interact with smartphones, and STVs are only used to present the final content. However, the contents of screen sharing in this method are limited, and the advantages of a large STV screen are not fully utilized. At the same time, older adults still face the challenges of using smartphones. For example, the small smartphone screen is hard to see clearly (Zhou, Rau, et al., 2014), and it is easy to unintentionally touch because users' fingers block the screen or older adults' finger flexibility is low (Chêne et al., 2016).

The third way is to allow users to focus on the STV and smartphone screen at the same time and provide solutions to reduce the attentional switches of screen mirroring. There are two possible solutions: increasing button sizes and improving tapping methods.

With regard to the button sizes, a lot of researchers focus on the effect of button sizes on touchscreens but few on the

effect of that on the screen mirroring. The researches of button sizes on touchscreens are summarized in Table 1. As for small touchscreens (i.e., smartphones and tablets; screen: 9.7 inches and below), the button size suitable for older adults should be at least 8 mm and is best in the range of 14–17.5 mm. As for large touchscreens (i.e., PDA, ATM, etc.; screen: above 9.7 inches), the suitable button size for older adults ranges from 15.9–20 mm; 20 mm is also recommended by a study involving younger adults (Colle & Hiszem, 2004) and a study involving disabled people (Chourasia et al., 2013). For screen mirroring, there is no direct design guideline for button sizes that are appropriate for older adults to provide to practitioners. Considering the frequent attentional switches in screen mirroring, it could be speculated that the suitable button size for older adults might be larger than those on small touchscreens and that the button sizes on large touchscreens would be more appropriate. Therefore, we investigated the performance and satisfaction of the button sizes ranging from 8 to 20 mm; in addition, a middle size, 14 mm, was selected. The hypotheses are as follows:

H1 & H2: When older adults use screen mirroring (mirroring smartphones screen to STVs), among the 8, 14, and 20 mm target button sizes on the smartphone screen, the 20 mm button size has the best tap performance (H1a) and satisfaction (H1b), while the 8 mm button size has the worst tap performance (H2a) and satisfaction (H2b).

With regard to the tapping methods, this study aimed to propose new touch gestures instead of the direct tap to improve the user experience of screen mirroring. The direct tap method is commonly used on touchscreens and needs users to look at the touchscreen and accurately determine the position of fingers. However, in the screen mirroring, every time users look down at the touchscreen, it causes a shift of attention, which may increase the risk of unintentional touch for older adults. In this study, there might be two possible improved touch gestures more suitable for the screen mirroring. First, the "Slide + Tap" method divides the gesture into sliding to select a target and then tapping the screen anywhere to confirm this selection; this has been suggested by Chêne et al. (2016) who used the Drag & Swipe two steps on touchscreens to help older adults reduce unintentional touch. However, the "Slide + Tap" method might increase the task completion time because of the increase in interaction steps. Second, the "Slide + Release" method only requires users to slide to select a target and then release their fingers to confirm this selection; this is referred to the study of Jung and Jang (2015) who designed the "Press and Flick" to relieve the unintentional touch of the hyperlink for younger adults (20–26 years old). Compared to the "Slide + Tap," the "Slide + Release" actually has only one step, which may lead to a shorter task completion time. The hypotheses are as follows:

H3: When older adults use screen mirroring (mirroring smartphones screen to STVs), the tap performance (H3a) and satisfaction (H3b) of the "Slide + Tap" are higher than that of the "Tap."

Table 1. Summary of researches about button sizes on touchscreens.

Device	Screen size	Age	touch gestures	Postures	Button size (mm)	Space (mm)	Time	Measures	Results	Reference
Small touchscreen	3.5in	60–80	Tap, drag, zoom	Two-hand held	30, 50, 70 pi [Single block]	Null			30-pixel targets performed worst and should be larger than 8 mm.	(Kobayashi et al., 2011)
	4.0in	65–95	Tap, swipe	Null	21, 17.5, 14, 10.5, 7 [Block array]	0, 3.5, 7, 10.5	Accuracy rates, task completion times, and subjective preferences		14 and 17.5 mm performed well.	(Leitão & Silva, 2012)
	4.3in	Above 65	Tap	Null	5, 8, 12 [Block array]	1, 3	task completion time, number of error, suitability, ease of use, mental effort, physical effort		The performance increased with target and spacing sizes.	(Hwangbo et al., 2013)
Large touchscreen	12.1in	18–22	Tap	No hand held	10, 15, 20, 25 [Block array]	1, 3	Response time and errors, and user preferences		20 mm was found to be sufficiently large.	(Colle & Hiszem, 2004)
	15in	People with disabilities (52.3), without (48.9)	Tap	No hand held	10, 15, 20, 25, 30 [Block array]	3, 5	Performance (accuracy and timing) and touch characteristics (force and impulse)		Standing affects performance at smaller button sizes (< 20 mm).	(Chourasia et al., 2013)
	17in	55–85	Tap	No hand held	11.4, 14.0, 16.5, 19.1, 21.6, 24.1 [Block array]	0, 3.2, 6.4, 12.7, 19.1	Reaction time, accuracy and user preferences		16.5–19.1 mm button size performed well.	(Jin et al., 2007)
	23in	52–81	Tap, drag, zoom, rotate	No hand held	5.3 × 3.0, 10.6 × 6.0, 15.9 × 9.0, 21.2 × 12.0	0, 1, 3, 6	Number of errors, time, and satisfaction		15.9 × 9.0 mm led to better performance and higher satisfaction.	(Gao & Sun, 2015)

H4: When older adults use screen mirroring (mirroring smartphones screen to STVs), the tap performance (H4a) and satisfaction (H4b) of the “Slide + Release” are higher than that of the “Tap.”

Input postures may affect the tap performance. There are two common input postures to use smartphones: two-handed and one-handed postures. The two-handed posture is commonly used when sitting in a seat or lying in bed with two hands free (Chen, 2014). By contrast, the one-handed posture is used when the user has only one hand free, for example, when holding the handrail with one hand on the bus. Between the two postures, the former is mainly operated with the index finger, while the latter mainly uses the thumb. However, there are differences in finger sizes, degrees of freedom, and movement abilities between the index finger and thumb (Zhang et al., 2005). For example, the interphalangeal joint of the thumb is wider than the distal joint of the index finger (Chinese Academy of Standardization and Information Classified Coding, 1996), but the degree of freedom of the thumb is greater than that of the index finger (Cash, 2011). Nevertheless, there is a lack of studies that consider the different input postures when discussing the effect of button sizes on the tap performance. A similar study indicated that younger adults who used the thumb had a better performance than who used the index finger, but there was no significant difference in satisfaction (Chen, 2014). The related hypothesis is as follows:

H5: When older adults use screen mirroring (mirroring smartphones screen to STVs), the tap performance with the two-handed posture is higher than with the one-handed posture.

3. Experiment I

An experiment was conducted to investigate the difficulties for older adults in using touch gestures (“Drag,” “Slide,” “Zoom,” “Draw,” and “Handwrite”) in screen mirroring. For further investigation, the eye-tracking and age effects were considered in this experiment.

3.1. Variables

The independent variables were touch gestures (within-subject) and user groups (between-subject). The user groups included older and younger adults. There were five touch gestures, including “Drag,” “Slide,” “Zoom,” “Draw,” and “Handwrite,” selected in Experiment I. Following these previous studies (Cichoń et al., 2013; Gao & Sun, 2015), the tapping, dragging, sliding, pinching, swiping, zooming, rotate, etc., were investigated for older adults, which were proved not too difficult for them to use. The touch gestures should be selected according to some possible tasks in screen mirroring, such as changing a element’s location, browsing a web page, sharing a photo, drawing a mark to emphasize, and inputting some text with users’ smartphones. The first three tasks involve three commonly used touch gestures: “Drag,” “Slide,”

and “Zoom” (Brudy et al., 2019). The last two involved more difficult touch gestures: “Draw,” and “Handwrite” (considering Chinese older adults are prone on using handwriting to input Chinese characters) (Chen, 2014), which were chosen to broaden the range of the available tasks for older adults and find critical problems as many as possible.

The dependent variables were performance and satisfaction. Performance had three measures: task completion rate, task completion time, and attention ratio (on the STV screen). Task completion was defined as a participant completing a task without assistance. The task completion time was automatically recorded by the experimental system, starting from the time when the “Start” button was pressed on the task start page and ending at the time when the “Complete” button was pressed on the task end page. The attention ratio was used to measure the participants’ concentration on the STV screen or the smartphone screen and was collected by the Ergoneers eyeglass eye-tracking system, which was equal to the total gaze time on the STV screen divided by the total time of the task. Satisfaction included overall satisfaction, a ranking of the task difficulty, and overall satisfaction of using screen mirroring or using only a smartphone. Overall satisfaction was measured by an After-Scenario Questionnaire (ASQ) with three items using 7-point Likert scales (Lewis, 1995). The ranking of the task difficulty was sorted by participants after completing all tasks. Furthermore, to compare the experience of the touch gestures in using screen mirroring and using only a smartphone, the participants were asked to score the satisfaction of these two use conditions on a 7-point Likert scale. In addition, eight covariates were considered: age, gender, education, smartphone experience, smartphone usage time, dominate hand, dominate finger, and preferred posture with a smartphone.

3.2. Participants

A total of 14 older participants (five males and nine females; 63–78 years old) and 14 young participants (five males and nine females, 20–26 years old) were recruited for the experiment. The older participants were from an urban area in Chongqing, and the younger participants were students from Chongqing University. All participants were able bodied without any limitations on their hand movement and had not taken any long-term intense exercise (e.g., playing basketball and badminton) before the experiment to avoid the impact of hand fatigue. Other details are shown in Table 2.

3.3. Prototypes and tasks

The experimental prototypes were developed with the Proto.io and run on the iOS system. Proto.io is a mobile application prototyping platform, developed by Labs Division of SNQ Digital (San Francisco Bay Area, West Coast, Western US), which was chosen because it provided prototypes that run smoothly and support a variety of touch gestures.

There were five experimental tasks, including “Drag,” “Slide,” “Zoom,” “Draw,” and “Handwrite.” The screenshots of the experimental system are shown in Figure 1. It must be noted that the “Tap” or other touch gestures that needed users

Table 2. Demographic information description.

Variables	Information of older adults	Information of younger adults
Age	69.9 years old (SD = 5.2)	23.6 years old (SD = 1.7)
Job	42.9% factory worker, 57.1% others	100% student
Education	42.9% primary school and below, 42.9% middle school, 14.3% high school	85.7% graduate students, 14.3% undergraduates
Smartphone experience	57.1% have no experience, 42.9% have experience for 2.8 years (SD = 1.5)	100% have experience for 8.6 years (SD = 3.0)
Smartphone usage time	0.9 h (SD = 1.3)	8.4 h (SD = 3.7)
Watching TV time	2.9 h (SD = 3.5)	0.5 h (SD = 0.8)
Commonly used TV types	57.1% cable TVs, 14.3% STVs, 14.3% Satellite TVs, 14.3% others	64.3% STVs, 28.6% cable TVs, 7.1% others
Dominate hand	100% right hand	100% right hand
Dominate finger with a smartphone	21.4% thumb, 78.6% index finger	64.3% thumb, 35.7% index finger
Preferred posture with a smartphone	21.4% one-hand held, 78.6% two-hand held	64.3% one-hand held, 35.7% two-hand held

to confirm the location of their fingers on the screen still existed in each task, which might have influenced the user experience. The standard steps for each task are described below.

The “Drag” task required users to drag the green rectangle to the green box for five trials. In each trial, the positions of the rectangle and box were randomly changed. Participants needed to confirm their finger’s position at least once in each trial; this occurred when the drag started.

The “Slide” task required users to slide through the web page to find the specified number and to choose it. There were three different web page sizes (3600, 7200, and 14400 pixels) to increase the difficulty in sliding. There were six trials in total. At least one confirmation of the finger’s position was required in each trial; this occurred when the participants selected the target at the end of the task.

The “Zoom” required users to zoom the small red box to coincide with the large green box for two trials. Participants had to confirm their fingers’ position at least twice in each trial; this occurred when the participants placed two fingers on the screen and moved the picture with a single finger. However, the difficulty in completely overlapping two boxes led to more than one confirmation.

The “Draw” required users to draw a circle, a tick, and a cross in the white tracks for two trials. The fingers’ position needed to be confirmed at least three times in each trial because the participants need to determine the appropriate starting points. As the number of adjustments increased, the number of confirmations increased.

The “Handwrite” required users to trigger the input box and then handwrite the appointed Chinese characters for two trials. The appointed characters in the tasks were chosen from the previous study to reduce the influence of the characters (Chen, 2014). The fingers’ position needed to be confirmed at least twice in each trial; this occurred when triggering the box and starting to write. The number of confirmations increased as the number of written strokes increased.



Figure 1. The screenshots of the experimental tasks of the screen mirroring (From left to right: "Drag," "Slide," "Zoom," "Draw," and "Handwrite").

3.4. Apparatus

The experimental equipment is shown in **Figure 2**. To ensure the screen mirroring between smartphone and STV to run smoothly, the screen of a smartphone (iPhone7; screen: 4.7 inches; resolution: 1334×750 ; OS: iOS 12; role: input device) was mirrored onto a laptop computer (Xiaomi; resolution: 1920×1080 ; OS: Windows; role: like a set-up box) by the Airplay through the AirSever Connect application. And a 60-inch STV (Sharp; screen: 60 inches; resolution: 1920×1080 ; role: display device) was used to display the computer screen via an external HDMI cable. The experiment was conducted in a quiet office, and the participants sat on a sofa 3 m away from the STV.

An Ergoneers eye-tracking system developed by Ergoneers GmbH of Germany was used to collect the attention ratio on the STV and smartphone screen. This system included Dikablis Professional Glasses and D-Lab (version 3.0). There are two eye cameras and one field camera set up on the glasses. The tracking frequency of the eye camera is 60 Hz per eye and the pupil tracking accuracy is 0.05° visual angle. As for the field camera, the resolution is 1920×1080 pixels (Full HD) and the aperture angle is from 40° to 90° . With the D-Lab, the eye-tracking data were collected and analyzed in the experiment. The markers in the field videos can be connected with Area of Interest (AOI) of the STV and smartphone screen dynamically to obtain the attention ratios on the

STV and smartphone screen. However, due to the small scope of smartphone screens and the limited aperture angle of the field camera, it was difficult to capture the users' attention on the smartphone screen accurately. Therefore, the attention ratio of the STV screen was used to measure users' concentration on the STV screen rather than that of the smartphone screen.

3.5. Procedure

The experiment took about 50 minutes. First, the participants were given the informed consent and background information questionnaires. Second, the experimenter helped participants to put on the eye tracker and then calibrated it. After this, the screen brightness of the smartphone was turned down and participants were told to mainly look at the STV screen to prevent them from only staring at the smartphone. Third, the participants were trained and then asked to complete five touch gesture tasks independently and fill out the ASQ scales after each task. To avoid the effects of dominate hand, finger, and posture, participants were asked to use two hands to hold the smartphone (left hand to hold and right hand to operate). Last, the participants were required to rank the difficulty of the five touch gestures and then participate in a brief interview. The interview required participants to evaluate their

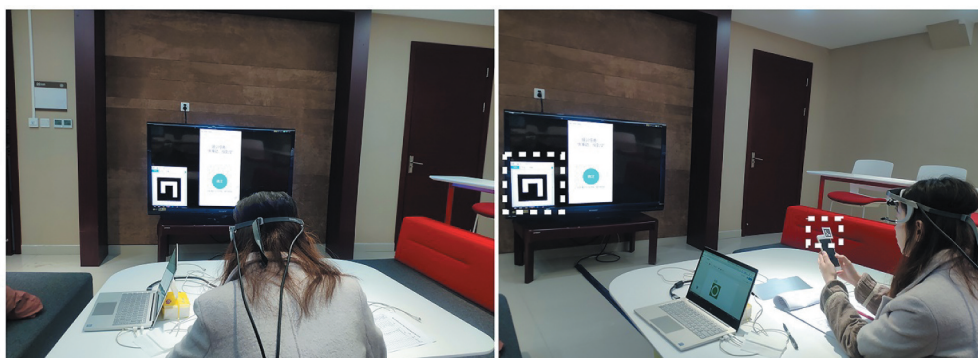


Figure 2. The experimental equipment and environment. Notes: the markers in the white dotted boxes were used to mark the area of interests (AOI) to analyze the eye-tracking data.

satisfactions of using the screen mirroring and using the smartphone screen only.

3.6. Results and discussion

3.6.1. Performance

In terms of task completion rate, older participants completed 77.9% of the tasks and younger participants completed 100% of the tasks. For the older participants, the relatively difficult touch gestures were “Zoom,” “Draw,” and “Handwrite,” with 57.1%, 64.3%, and 67.9% successful tasks. In terms of task completion time, since the task completion time of the failed tasks was invalid (about 5.5%), these data were deleted before the covariance analysis was conducted. After deleting the invalid data, the older participants spent an average of 36.8 s per task (SD = 2.2) and the younger participants spent 18.5 s per task (SD = 1.2).

The correlation analysis, ANCOVA, and paired t-test were used to analyze the performance. First, all eight covariates were included in the ANCOVA model. Then, the covariates with a regression slope equal to 0 and those without significant influence in the ANCOVA were removed; the remaining covariates will be reported separately. Since Mauchly's test of sphericity in the ANOVA models was violated, a Huynh-Feldt correction was used. Furthermore, if the effects of the independent variables were significant, the paired sample t-test method would be used to further analyze the impact of different levels on the dependent variable.

As Table 3 and Figure 3 show, there were two interaction effects of touch gestures and user groups on task completion rate and task completion time. In particular, older participants

took a significantly longer time than younger participants, especially for the “Zoom” task, using 72.2% more time than younger participants. One possible reason was that the “Zoom” involved multiple fingers and it is a challenge for older adults to switch between fingers.

According to the results of the performance, the difficult touch gestures in screen mirroring for older adults were “Zoom,” “Draw,” and “Handwrite,” and the “Drag” and “Slide” were relatively easier. The reason for the difficulty of these touch gestures in screen mirroring may be that users have to frequently switch their attentions to confirm their fingers' position when tapping. The more confirmations, the more times users switched their attention between the STV screen and the smartphone screen.

3.6.2. Satisfaction

A two-way ANCOVA of user group and touch gesture was conducted in terms of overall satisfaction (ASQ scores), and the results indicated the main effects of user group ($F(1, 26) = 13.843, p = .001, \eta_p^2 = 0.347$) and touch gesture ($F(1, 26) = 10.172, p = .000, \eta_p^2 = 0.281$). As shown in Figure 4, older participants gave a higher satisfaction score ($M = 16.4, SD = 3.5$) than younger participants ($M = 14.5, SD = 5.2$). The total satisfaction score was 21 points. Regardless of age, the overall satisfaction of “Zoom” and “Draw” tasks was the lowest and that of the “Drag” was the highest. Similarly, the results of the ranking of the task difficulty indicated that the “Draw” and “Zoom” tasks were more difficult and the “Drag” task was easier for both older and younger participants (see Figure 5).

In terms of the overall satisfaction of using the touch gestures in two use conditions (in the screen mirroring and on the smartphone only), the results showed that the touch gestures and the use conditions had a significant interaction effect ($F(1, 52) = 12.968, p = .000, \eta_p^2 = 0.200$) as Figure 6 illustrates. The “Drag” had a 21.6% higher score when being operated in screen mirroring than on the smartphone only ($T = 4.86, p = .000$). One possible reason may be that the screen of the STV was bigger to ensure users saw the interface clearly than that of the smartphone. However, as the difficulty of touch gestures

Table 3. The effects of the touch gestures and age groups on performance.

Variable	Task completion rate	Task completion time
Touch gesture	$F(1,25) = 0.595, p = .612, \eta_p^2 = 0.023$	$F(1,33) = 2.604, p = .076, \eta_p^2 = 0.073$
User group	$F(1,25) = 35.422, p = .000, \eta_p^2 = 0.586$	$F(1,33) = 5.534, p = .025, \eta_p^2 = 0.144$
Touch gesture $\times$ user group	$F(1,25) = 6.118, p = .001, \eta_p^2 = 0.197$	$F(1,33) = 3.391, p = .035, \eta_p^2 = 0.093$

Note. η_p^2 = Partial η^2 , *Significant at 0.05 level.

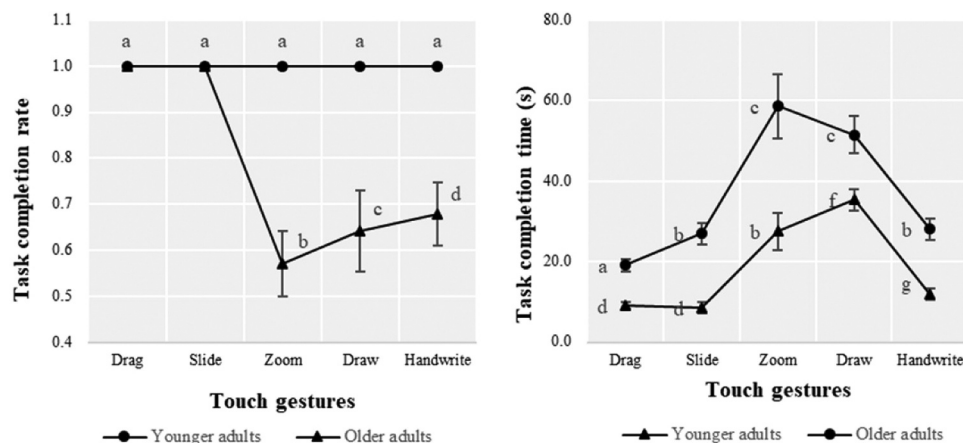


Figure 3. The task completion rate (left) and task completion time (right) of age groups. Note: The levels that differed significantly at $p < .05$ are marked with identical symbols (a, b, c, d, e, f, g); error bars: standard error.

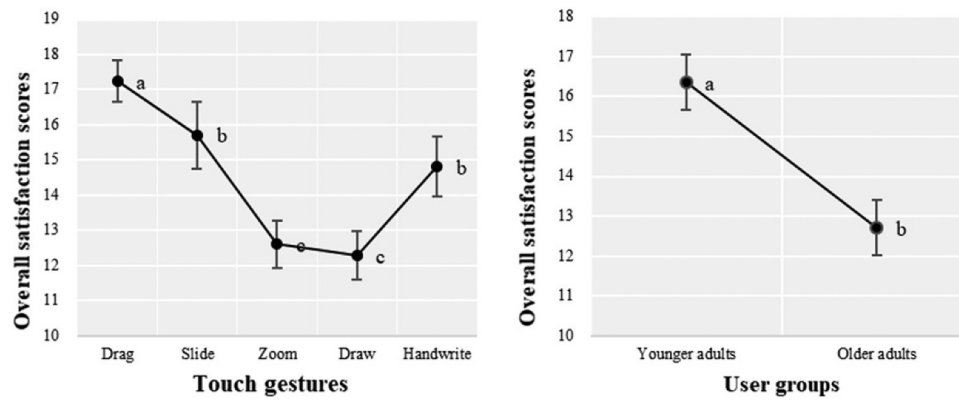


Figure 4. The overall satisfaction scores on the touch gestures [left] and users groups [right] of the screen mirroring.

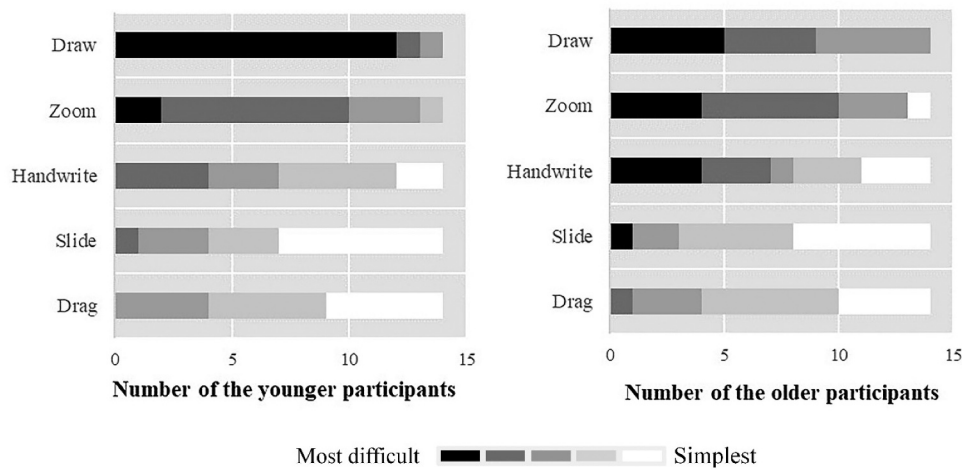


Figure 5. The difficulty ranking of the five touch gestures for younger participants (left) and older participants (right).

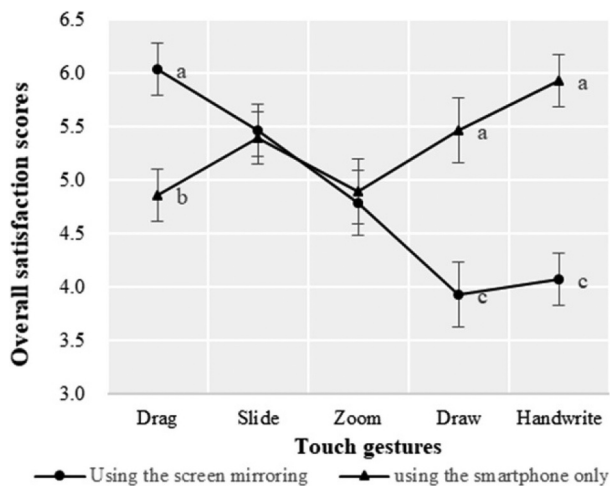


Figure 6. The effect of the touch gestures and use conditions on overall satisfaction scores.

increased, the advantages of big STV screens decreased. Specifically, the satisfaction achieved by using the smartphone only was 32.7% higher ($T = -5.06, p = .000$) than that by using screen mirroring when completing the “Draw” task, and 37.1% higher ($T = -7.62, p = .000$) when completing the “Handwrite” task.

To further reveal the critical problem in screen mirroring, the interview was analyzed. As shown in Table 4, for the older participants, the advantage of screen mirroring was that the STV screen was bigger and clearer than that of smartphone only. However, when completing the complex touch gestures (e.g., zoom and draw), the tasks became too difficult to allow users to enjoy the advantage of the big STV screen. In particular, the most frequently mentioned problem was that “I had to look down at the smartphone to confirm their fingers location,” both for older and younger participants.

3.6.3. Attention ratio

Two points of calculating the attention ratios need to be noted. First, a gaze behavior with a gaze time of less than 120 ms was deleted in this calculation because it was too short to be regarded as a proper glance according to the D-Lab Manual (version 3.0). Second, the captured attention ratios on the smartphone screen might be much lower than the actual assigned attention of participants because participants’ pupils might have been blocked by their eyelids when they looked down at the smartphone. In contrast, the attention ratio on the STV screen was a better index to measure users’ concentration on the STV screen.

On average, there was 44.4% (SD = 0.04) attention focused on the STV screen and 9.4% (SD = 0.04) on the smartphone

Table 4. The results of the interview.

Use conditions	Types	Feedback	No. Y	No. O
Screen mirroring	Likes	The STVs suit for sharing.	12	2
		The STV screen is big to see clearly.	6	5
		Prefer the STVs more than smartphone.	2	3
		The STV screen has higher brightness.	1	1
		It was easy to handwrite looking at the STV screen.	2	No
	Dislikes	The text on the STV screen is clearly.	1	No
		It was preferred to watch the STV and smartphone both.	No	1
		Users had to look down at the smartphone to confirm their fingers location.	14	3
		Screen segmentation lead to unsmooth, unsynchronized experience.	5	2
		It was hard to handwrite well when looking at the STV screen.	2	1
Smartphone only	Likes	The STV was too far away.	1	No
		Users felt nervous when using screen mirroring.	1	No
	Dislikes	The smartphone were felt more convenient, faster, better to control than the STV.	4	10
		It was familiar to use smartphone only rather than screen mirroring.	5	3
		The smartphone screen was close enough to see clearly.	3	3
		Difficult or time-consuming touch gestures should be done with a smartphone only to save time.	8	No
	Dislikes	The smartphone screen was too small to be covered by users' fingers.	6	1
		The smartphone screen was too small to complete difficult gestures (e.g., zoom and draw).	1	3
		Older adults with poor eyesight can't see the smartphone clearly.	No	3
		Sliding on smartphone screen was too fast.	No	1
Other problems		Users didn't learned how to handwrite.	No	3
		The "Zoom" task was difficult.	No	2
		The "Slide" on the STV screen was not smooth.	1	1
		The "Draw" task was difficult.	1	No

Notes: No. Y = Number of younger participants, No. O = Number of older participants.

screen by participants. The means of the attention ratios (on the STV screen) of different touch gestures for the two user groups were compared (see Figure 7). Younger adults had higher concentration on the STV screen when they were dragging and sliding, while their attentional switches

increased when completing other touch gestures. For the older participants, except for the "Drag" task, their attentional switches in screen mirroring when completing other touch gestures were not significantly different.

To investigate whether the attention ratio (STV screen) had an impact on performance and satisfaction, correlation analyses were conducted. As a result, the attention ratio (STV screen) was significantly related to the task completion time ($r = -0.187, p = .000$). In other words, the more concentration participants gave to the STV screen, the shorter the task completion time; this was consistent with Rashid et al. (2012a) who found the average time of an attention switch in using hybrid UI configurations (e.g., using a smartphone and TV at the same time) was 1.8 s and pointed out that the critical problem that decreased user experiences was the frequent attentional switches.

3.6.4. Effects of the covariates

Among eight covariates, the remaining three covariates (i.e., age, education, and gender) after conducting the ANCOVA were reported (see Table 5). The touch gestures and ages had a significant interaction effect on the task completion time. Among these, the "Zoom" was influenced by ages; when the users' ages increased, the task completion time of the "Zoom" task rapidly increased. Besides, the results of the correlation analysis showed that age was positively related ($r = 0.402, p = .000$) with task completion time and education was negatively related ($r = -0.418, p = .000$) with task completion time, which means that the older the age, and the lower the education, the worse the performance of these touch gestures. Additionally, the results showed that the males' completion rate was 13.84% higher ($t = 5.876, p = .000$) than the females'.

4. Experiment II

According to the results of experiment I, the major problem with the screen mirroring for older adults was that they had to frequently switch their attentions between the smartphone and the STV screen, especially when tapping on the touchscreen. Therefore, experiment II focused on the "Tap" task and proposed solutions to the frequent attentional switches of the screen mirroring.

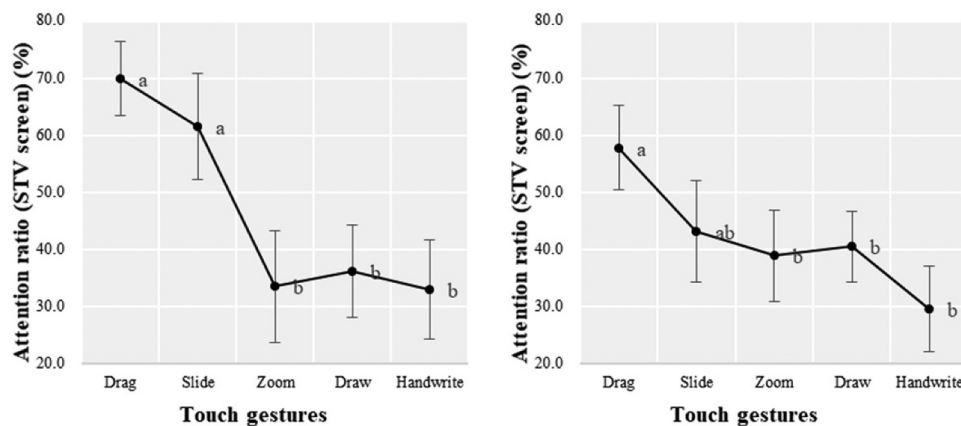


Figure 7. The AOL attention ratios (STV screen) of five touch gestures of younger participants (left) and older participants (right).

Table 5. The effects of covariates on performance.

Dependent variables	Covariates	Effects
Task completion rate	gender	$F_{(1,25)} = 10.459, p = .003^{**}, \eta_p^2 = 0.295$
Task completion time	Education	$F_{(1,33)} = 15.759, p = .000^{**}, \eta_p^2 = 0.323$
	Age	$F_{(1,33)} = 6.172, p = .018^*, \eta_p^2 = 0.158$
	Touch gesture $\times$ Age	$F_{(1,33)} = 3.556, p = .030^*, \eta_p^2 = 0.097$

Note. η_p^2 = Partial η^2 , **Significant at the 0.01 level, *Significant at the 0.05 level.

4.1. Variables

The independent variables were the tap methods, button sizes, and input postures. The tap methods had three levels: “Tap,” “Slide + Tap,” and “Slide + Release.” The button sizes included three levels: 8 mm, 14 mm, and 20 mm. Both variables were within-subject. There were two levels of input postures (between-subject): two-handed and one-handed postures.

The dependent variables were performance and satisfaction. The performance was measured by three dimensions, including task completion rate, task completion time, and attention ratio (STV screen), which were the same as experiment I. Satisfaction included perceived ease of use, attention to use, and overall satisfaction. The perceived ease of use and attention to use were measured by the Post-Experiment System Availability Questionnaire (PSSUQ) with 7-point Likert scales (Lewis, 1995; Zhou, Zhang, et al., 2014). Additionally, the overall satisfaction was subjectively rated by participants via a one-question questionnaire with a 7-point Likert scale (Sauro & Dumas, 2009; Tedesco & Tullis, 2006). As for the covariates, besides the eight covariates included in experiment I, finger dexterity was added because it might affect the time of tapping for older adults.

4.2. Participants

A total of 30 older participants (eight males and 22 females) were recruited for the experiment. The participants were from an urban area in Chongqing, China. The participants were in good health, except for one participant who reported having Parkinson’s syndrome. Although this participant had a tremor in his hands, he had no difficulty in performing the tasks independently. Other detailed information of the participants is presented in Table 6.

Table 6. Demographic information description.

Variables	Information of older adults
Age	66.3 years old (SD = 3.7)
Job	80% physical labors, 20% mental labors
Education	33.3% primary school and below, 46.7% middle school, 13.3% high school, 6.7% undergraduate
Smartphone experience	26.7% have no experience, 3.3% have experience for 3.4 years (SD = 2.0)
Smartphone usage time	3.4 h (SD = 2.3)
Watching TV time	86.8% watching TV for 2.1 h (SD = 1.7) per day
Commonly used TV types	70% cable TVs, 30% STVs
Dominate hand	93.3% right hand, 6.7% left hand
Dominate finger with a smartphone	23.3% thumb, 73.3% index finger, 3.3% middle finger
Preferred posture with a smartphone	20% one-hand held, 80% two-hand held

4.3. Apparatus

The experimental system with 3 (button sizes) $\times$ 3 (tap methods) types of interfaces was developed with Android 5.1 as shown in Figures 8 and 9. Each interface contained six buttons in the middle of the screen with a gap of 5 mm between elements, and the buttons were numbered from 1 to 6 randomly without repetition. As for the tap methods, the “Tap” method is no different from that on smartphone touchscreen. While the improved “Slide + Tap” method has two steps to complete a tap (slide the finger to focus a target then release and tap anywhere to confirm), and the “Slide + Release” method has only one step (slide the finger to focus a target then release to confirm). In both improved methods, a fingerprint was used to indicate the current location of the user’s finger.

The equipment used in experiment I was also involved in experiment II (see Figure 10), including the eye tracker. In addition, a finger dexterity tester was used to measure participants’ finger dexterity by recording the total time spent by the participants to place iron needles into small pinholes. The way to mirror the smartphone screen to the STV was similar as experiment I, except that the smartphone was changed to an Android phone (Huawei; screen: 5.2 inches; resolution: 1080 $\times$ 1920; OS: Android 6.0) and the mirroring application was changed to the ApowerMirror application (Room 345, third Floor, Peninsula Center, 67 Mody Road, Tsim Sha Tsui, Kowloon, Hong Kong) in order to meet technical requirements as well as ensure the experimental system run smoothly. Given that data of the two experiments were not compared to draw conclusions, the influence of devices and applications might not be significant.

4.4. Procedure

The experiment took about 30 minutes. First, the participants were asked to fill in the informed consent form and background information questionnaire. Second, the participants were required to complete the finger dexterity test. Third, the experimenter helped participants to put on the eye tracker and calibrated it. Then, participants were given the necessary training before the formal tasks. After that, participants were randomly asked to use either the one-handed posture (operating with their thumbs) or the two-handed posture (operating with their index fingers) to complete the tasks, and fill in the QSSUQ questionnaire after each task (Lewis, 1995). There were 27 tasks (3 button sizes $\times$ 3 tap methods $\times$ 3 trials), which were to tap the randomly appointed button correctly. The task would be recorded as a success if the button was tapped correctly; otherwise, it was recorded as a failure. The



Figure 8. The screenshots of the prototypes with 8 (left), 14 (center), and 20 (right) mm buttons on the smartphone screen.

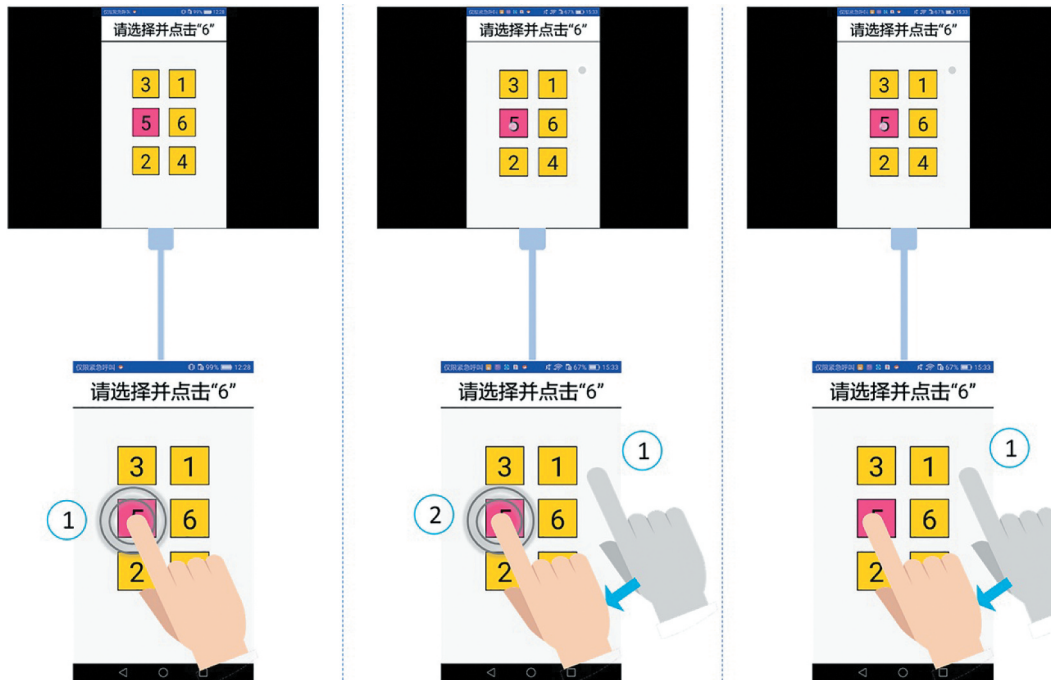


Figure 9. The schematic diagrams of "Tap" method (left), "Slide + Tap" method (center), and "Slide + Release" method (right). Note: the above parts are the STV screens and the bottom parts are the corresponding smartphone screens. The numbers on the diagrams indicate the interactive steps of the touch gestures.

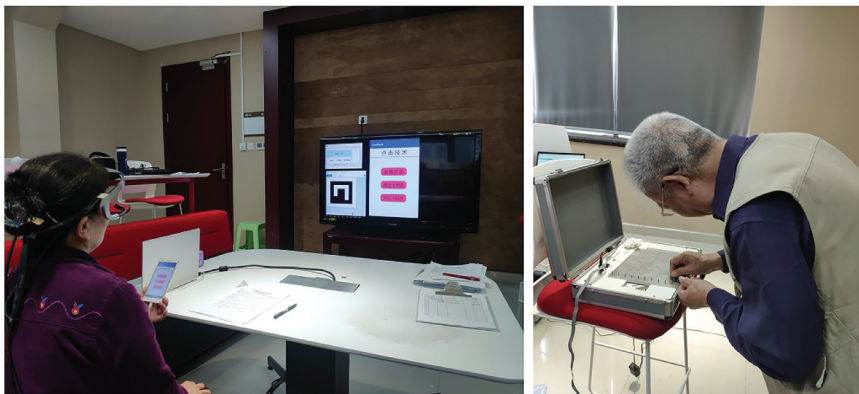


Figure 10. The experimental hardware and environment (Left: the formal experiment; Right: before the formal experiment).

task order was counterbalanced to remove the order effect. Finally, participants rated their overall satisfaction with the touch gestures and button sizes and were required to participate in a brief interview. Three questions were asked: “What do you think are the advantages and disadvantages of the three touch gestures?” “How annoyed by or how much did they accept the attentional switches?”

4.5. Results and discussion

4.5.1. Performance

Overall, the participants completed 97% of the tasks ($SD = 0.2$) and spent 5.0 s ($SD = 6.0$) per task. On average, an attention ratio of 61.4% ($SD = 0.23$) per task was detected according to the eye-tracking data. The attention ratio for the “Tap” task was 49.0% ($SD = 0.19$), and those for the improved groups (“Slide + Tap” and “Slide + Release”) were 68.0% ($SD = 0.23$) and 66.8% ($SD = 0.21$).

The data analysis methods of this experiment were the same as experiment I. It should be noted that the eye-tracking data of one participant using the one-handed posture and one using the two-handed posture were mostly lost. To ensure the reliability of the data, the eye-tracking data of the two participants were deleted.

The interactive effects of button sizes and input postures on task completion rate and completion time are shown in Table 7 and Figure 11. As for different input postures, when using the one-handed posture, compared with the 14 mm button, using the 8 mm button had an 8.6% ($t = -3.95$, $p = .000$) lower task completion rate and a 40.7% ($t = 4.46$, $p =$

.000) higher completion time. However, there was no significant difference between the three button sizes when using the two-handed posture. As for different button sizes, when using the button size of 8 mm, using the two-handed posture can save 48.1% ($t = 4.03$, $p = .000$) of time compared to using the one-handed posture. Apart from the interaction effects, the main effects were also identified. As for the tap methods, the attention ratios (STV screen) when using the “Slide + Tap” and “Slide + Release” method were significantly higher than when using tap by 30.3% ($t = -4.86$, $p = .000$) and 28.3% ($t = 4.66$, $p = .00$). Therefore,

H2a, H3a, H4a, and H5 are partially supported, while H1a is not.

In addition, education ($F(1,68) = 16.921$, $p = .000$, $\eta_p^2 = 0.199$) and finger dexterity ($F(1,68) = 17.364$, $p = .000$, $\eta_p^2 = 0.203$) had a significant impact on the task completion time. Specifically, education was negatively ($r = -0.195$, $p = .000$) and finger dexterity was positively ($r = 0.192$, $P = .000$) correlated with task completion time, which means that the higher the education and finger dexterity, the shorter the task completion time; this is consistent with the empirical evidence. However, the user performance in this experiment was not significantly influenced by other covariates.

4.5.2. Satisfaction

Basic satisfaction data were computed. Of a total score of 7, the older participants scored 6.1 ($SD = 0.9$), 5.1 ($SD = 1.4$), and 5.6 ($SD = 1.1$) for “Tap,” “Slide + Tap,” and “Slide +

Table 7. The effects of touch gestures, button sizes, and input postures on user performance.

Variables	Task completion rate	Task completion time	Attention ratio (STV screen)
TM	$F_{(1, 88)} = 1.933$, $p = .157$, $\eta_p^2 = 0.021$	$F_{(1, 68)} = 1.034$, $p = .358$, $\eta_p^2 = 0.015$	$F_{(1, 25)} = 10.319$, $p = .000^{**}$, $\eta_p^2 = 0.292$
BS	$F_{(1, 88)} = 4.125$, $p = .025^*$, $\eta_p^2 = 0.045$	$F_{(1, 68)} = 0.373$, $p = .593$, $\eta_p^2 = 0.005$	$F_{(1, 25)} = 0.277$, $p = .759$, $\eta_p^2 = 0.011$
IP	$F_{(1, 88)} = 5.446$, $p = .022^*$, $\eta_p^2 = 0.580$	$F_{(1, 68)} = 9.260$, $p = .003^{**}$, $\eta_p^2 = 0.120$	$F_{(1, 25)} = 0.412$, $p = .527$, $\eta_p^2 = 0.016$
TM $\times$ BS	$F_{(1, 88)} = 0.848$, $p = .480$, $\eta_p^2 = 0.010$	$F_{(1, 68)} = 0.846$, $p = .412$, $\eta_p^2 = 0.012$	$F_{(1, 25)} = 0.256$, $p = .905$, $\eta_p^2 = 0.010$
TM $\times$ IP	$F_{(1, 88)} = 1.197$, $p = .298$, $\eta_p^2 = 0.013$	$F_{(1, 68)} = 0.621$, $p = .537$, $\eta_p^2 = 0.009$	$F_{(1, 25)} = 2.506$, $p = .092$, $\eta_p^2 = 0.091$
BS $\times$ IP	$F_{(1, 88)} = 4.125$, $p = .025^*$, $\eta_p^2 = 0.045$	$F_{(1, 68)} = 5.639$, $p = .014^*$, $\eta_p^2 = 0.077$	$F_{(1, 25)} = 0.562$, $p = .573$, $\eta_p^2 = 0.022$
TM $\times$ BS $\times$ IP	$F_{(1, 88)} = 0.141$, $p = .950$, $\eta_p^2 = 0.002$	$F_{(1, 68)} = 0.453$, $p = .599$, $\eta_p^2 = 0.007$	$F_{(1, 25)} = 0.524$, $p = .718$, $\eta_p^2 = 0.021$

Note. TM = Tap Method, BS = Button Size, IP = Input Posture, η_p^2 = Partial η^2 , *Significant at 0.05 level; **Significant at 0.01 level.

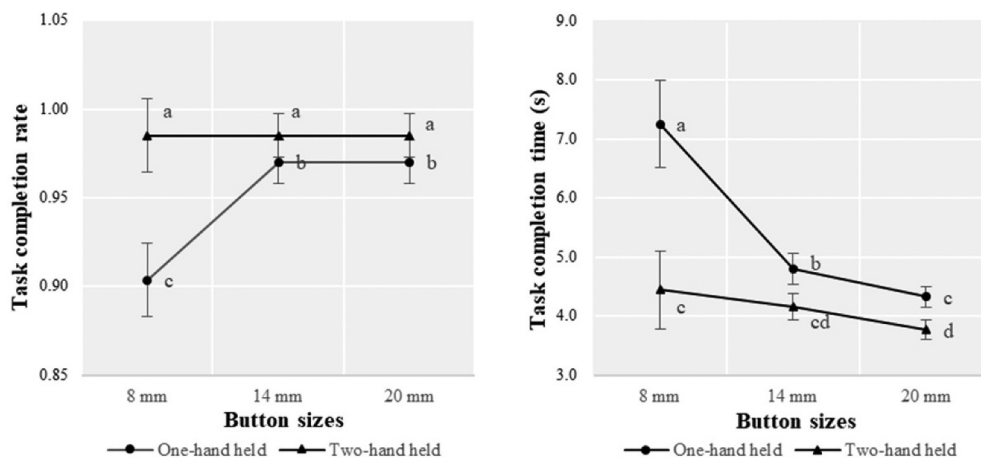


Figure 11. The effects of touch gestures, button sizes, and input postures on user performance.

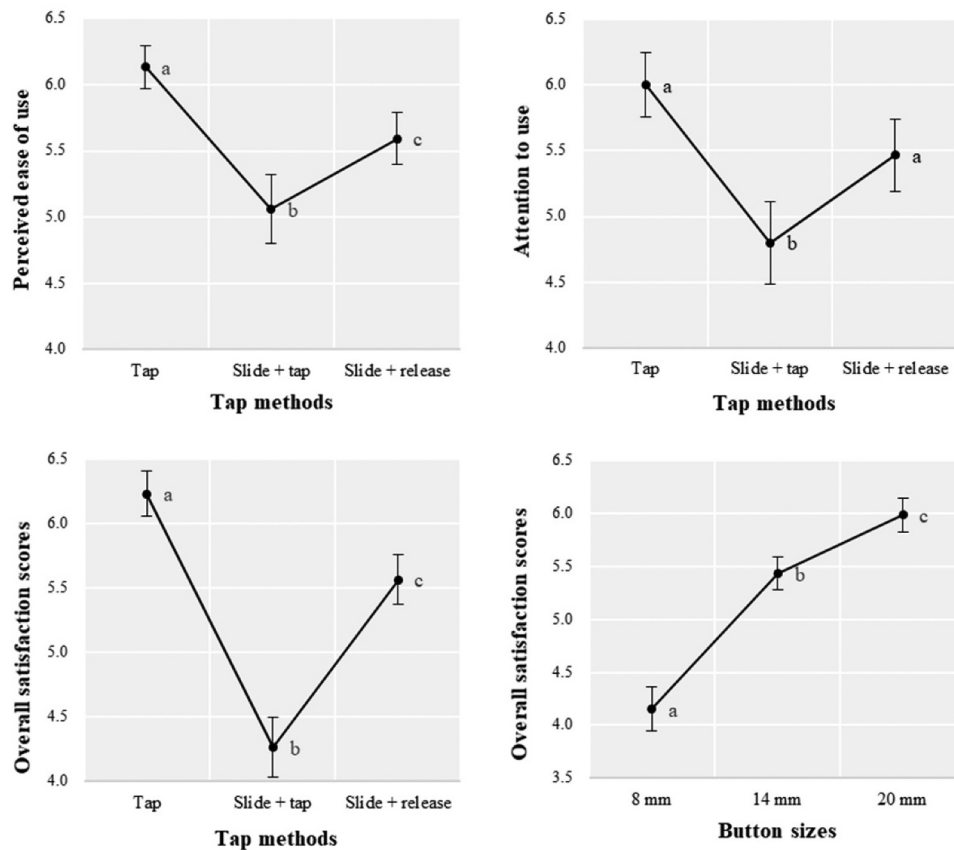


Figure 12. The effects of touch gestures on perceived ease of use (top left), attention to use (top right), and overall satisfaction (bottom left), and the effect of button sizes on overall satisfaction (bottom right).

Release,” on perceived ease of use and scored 6.0 (SD = 1.4), 4.8 (SD = 1.8), and 5.5 (SD = 1.5) on attention to use.

The ANCOVAs for satisfaction were conducted and the main effects were detected (see Figure 12). As for the tap methods, it had a significant impact on perceived ease of use ($F(1,28) = 13.776, p = .000, \eta_p^2 = 0.330$), willingness to use ($F(1,28) = 5.770, p = .005, \eta_p^2 = 0.171$), and overall satisfaction ($F(1,28) = 42.922, p = .000, \eta_p^2 = .605$). Regardless of the button sizes and input postures, the “Slide + Tap” method had the lowest perceived ease of use, attention to use, and overall satisfaction, while the “Tap” method obtained the highest scores. As for the button sizes, the overall satisfaction of the three button sizes was significantly different ($F(1,28) = 63.375, p = .000, \eta_p^2 = 0.694$). In screen mirroring, the larger the button size on the smartphone, the higher the overall satisfaction. In particular, the overall satisfaction of the 14 mm button size was 26.7% higher than that of the 8 mm ($T = -6.930, p = .000$). However, the input posture had no significant effect on the satisfaction. Therefore,

H1b and H2b are partially supported, while H3b and H4b are not supported.

To further reveal participants’ opinions about the three touch gestures, the interview results were analyzed next. The keywords were extracted and counted from the feedback of participants on the advantages and disadvantages of the three

touch gestures (see Table 8). From this table, it can be seen that participants found “Tap” to be simpler, faster, and more convenient, but in order to tap correctly, they had to have their eyes on the smartphone screen. Meanwhile, the problem of too many steps in “Slide + Tap” was most serious. Furthermore, 63.3% of the participants thought that it was better to have less attention switches, especially for users with neck pain, while 36.7% thought that the attention switches were acceptable.

5. General discussion

5.1. Major findings

This study sought to investigate the critical problem for older adults in interacting with screen mirroring between an STV and a smartphone and offer solutions to this problem. The results showed that the difficult touch gestures would hinder the user experience of the screen mirroring for older adults. Specifically, except for the “Drag” gesture, the performance and satisfaction of using “Slide,” “Zoom,” “Draw,” and “Handwrite” in screen mirroring were worse than using only on a smartphone screen. One critical problem making these touch gestures difficult in screen mirroring was that users had to frequently switch their attention between the STV and smartphone, which was also mentioned in the previous studies (Bellino, 2016; Dezfali et al., 2014; Rashid et al., 2012a; Sun et al., 2015). Considering that the “Tap” gesture was the foundation of other touch gestures and led to the obvious

Table 8. The feedback on three tap method from participants.

Tap methods	Advantages		Disadvantages	
	Keywords	Num.	Keywords	Num.
Tap	Simple	10	Easy to tap wrongly	5
	Fast	7	Have to gaze at the smartphone screen	5
	Convenient	5		
	Direct	4		
	Familiar	3		
Slide + Tap	Accurate	2	Troublesome	18
	Familiar	2	Slow	6
Slide + Release			Inconvenient	1
	Convenient	4	Slow	5
	Easy to adjust the position of fingers	3	Requiring high Finger dexterity	5
			Troublesome	4
	Accurate	3	Unfamiliar	2
	Simple	3		
	Fast	1		
	Comfortable	1		

attentional switches due to the need for accurate positioning, the top priority to improve the user experience of screen mirroring is to relieve the problem of frequent attentional switches when tapping.

Using improved tap methods was an effective way to relieve the attentional switches of screen mirroring. Older adults using the improved touch gestures (“Slide + Tap” and “Slide + Release”) had significantly higher attention ratios (68.0% and 66.8%) than using the normal “Tap” (49.0%). Furthermore, the improved tap method did not significantly harm the user performance even when the operation steps were increased in terms of task success and time consumption. Affected by users’ habits, the “Tap” method obtained the highest satisfaction, and the “Slide + Release” method obtained relatively high satisfaction scores. However, increasing the size of the buttons from 8 mm to 20 mm did not play a part in reducing the attentional switches in screen mirroring for older adults.

It should be noted that this result of the “Slide + Tap” was quite different from the previous studies. In this experiment, the “Slide + Tap” method received the worst feedback, whereas the previous studies suggested that it was useful for older adults to divide the “Tap” into two steps (i.e., selecting and confirming, Chêne et al., 2016; Jung & Jang, 2015; Liu & Zhou, 2018). There may be two reasons for this difference. First, the use scenarios were different. Liu and Zhou (2018) research focused on learning scenario instead of daily use, which emphasized the learning effect other than performance. Second, the purposes of use were different. Jung and Jang (2015) purpose was to alleviate the problem of accidental touch when users tapped the hyperlinks in web navigation (Jung & Jang, 2015). Conversely, the targets in our experiment were buttons, and they were not easily touched by mistakes. Therefore, when tapping the buttons on smartphones in the daily use of screen mirroring, the “Slide + Tap” method is not recommended for use when designing a screen-mirroring interface for older adults.

The input posture had an effect on the tap performance in screen mirroring. Using the one-handed posture had a lower task completion rate and longer task completion time than using the two-handed posture, which may result from the differences of the fingers for two aspects. First, the thumb mainly used by the one-handed posture is generally shorter and wider than the index finger mainly used by the two-

handed posture (Chinese Academy of Standardization and Information Classified Coding, 1996); thus, the motion capability of the thumb is lower than index finger and the response time is also longer (Chen, 2014). Second, when using the one-handed posture, the user’s range of motion is limited to one corner of the device screen (bottom left or bottom right) (Karlson et al., 2006). While the two-handed posture involves the fingers, wrists, and arms, making the range of motion more flexible than that of the one-handed posture.

Increasing the button size improved the performance and satisfaction when older adults used the one-handed posture to tap in screen mirroring. The smallest button size (8 mm) with the one-handed posture had the worst performance and satisfaction. In particular, the task completion rate using the 8 mm button size was 8.6% lower and task completion time was 40.7% higher than when using 14 mm with a one-handed posture. In a similar study, the researchers pointed out that when older adults used smartphones to complete the tap tasks, the task completion time when tapping the 12 mm buttons was 13.0% lower than when using 8 mm buttons (Hwangbo et al., 2013), which may mean that tapping in screen mirroring was more difficult than only on a smartphone. However, using the 20 mm button was not significantly different from using the 14 mm button in terms of performance, which meant that there is no need to use too large buttons in screen mirroring.

5.2. Practical implications

For practitioners, it is better to take care of the attentional switches between an STV and a smartphone in screen mirroring for older adults. Frequent attentional switches increase attention consumption and interaction difficulty of the touch gestures in screen mirroring. Specifically speaking, it is recommended that screen-mirroring interfaces are designed to avoid these difficult touch gestures as much as possible, such as the “Zoom,” “Draw,” and “Handwrite” tasks. More importantly, it is essential to provide solutions to reduce the difficulty of touch gestures in screen mirroring to allow older adults to fully enjoy the advantages of STVs with a large screen. Starting from the most basic aspect, improving the basic tap gesture is a promising solution.

The “Slide + Release” tap method is recommended to be added into the design of screen-mirroring applications instead of only providing the “Tap” method. The “Slide + Tap” and “Slide + Release” were both able to alleviate the problem of attentional switches in screen mirroring for older adults. Although the subjective feedback for these touch gestures was worse than that of the “Tap” method because of its familiarity and intuition, the “Slide + Release” method was acceptable when weighing both the performance and satisfaction. Considering that the “Slide + Release” method was learned for the first time in this experiment, it might be able to become preferred after continuous use. Thus, to maximize the user experience of the screen-mirroring application, there is a need to provide both the “Tap” and “Slide + Release” methods for users and provide operation instructions for users who are using “Slide + Release” for the first time.

In screen mirroring, using one hand to hold the devices and tap buttons caused performance reduction for older adults, and there might be two ways to solve this problem. The first way is to change from one hand to two hands. However, the use of postures is determined by the users’ personal habits and usage scenarios. In some cases, users have only one hand free, for example, when holding a pen, operating a mouse, turning over books in a meeting, or holding chopsticks or a soup spoon at home. Previous surveys suggest that most smartphone or PDA users prefer to use one hand to input because then they are able to handle other tasks at the same time (Karlson et al., 2006). Therefore, requiring users to input with two hands may not always be a good solution to improve the tap performance in screen mirroring. The second way is to adjust the interface design. Increasing the button size is a promising way to improve the performance and satisfaction when users use the one-handed posture. It is best to avoid using small buttons (smaller than 8 mm) when designing screen-mirroring applications for older adults; buttons larger than 14 mm are recommended.

5.3. Limitations and future studies

Four limitations of this study are noted. First, certain gestures used in this experiment are abstract and challenging for older adults, which might apply to a boarder population as well. This may lead to limitations in generalizing these design recommendations to realistic tasks. Second, due to technical limitations, this study did not use the same smartphones and operating system for the test of prototypes. Third, the experiments in this study only measured the initial use of the touch gestures and button sizes by older adults. It remains to be explored whether continuous use has different effects on these solutions when users tap the buttons in screen mirroring. Fourth, considering the importance and scope of the problem, this study only investigated tapping tasks. However, it is possible that the solutions in this study might have a negative impact on other types of tasks. For example, “Slide + Release” may conflict with the existing sliding gestures. Future work may be expanded to explore the effect of the “Slide + Release” method on other touch gestures and to provide improved schemes for other common touch gestures in screen mirroring.

6. Conclusion

This study aimed to investigate the difficulties about touch gestures of screen mirroring for older adults through Experiment I and to find solutions to the major problems (i.e., the frequent attentional switches between the STV and smartphone screens when tapping) via Experiment II. This study then attempted to solve this problem by improving the touch gestures (“Slide + Tap” and “Slide + Release”) and increasing the button sizes (8, 14, and 20 mm).

There were four findings. First, using the “Zoom,” “Draw,” and “Handwrite” gestures in screen mirroring was difficult for older adults; these had task completion rates lower than 68%. Second, the major problem of the screen mirroring for older adults was the frequent attentional switches between the STV and smartphone screens, which became obvious when tapping on the screen. Third, the improved “Slide + Tap” and “Slide + Release” touch gestures led to higher attention concentration (on STV screen) of screen mirroring by 30.3% and 28.3% than the “Tap” for older adults when completing tap tasks. Considering the “Slide + Tap” obtained the worst satisfaction, it is recommended to provide both the “Tap” and “Slide + Release” methods for older adults to choose in screen-mirroring applications. Fourth, the tap performance with the one-handed posture in screen mirroring for older adults exhibited 3.8% lower task completion rate and 27.9% longer time than that with the two-handed posture, which can be improved by increasing the button sizes. In particular, the 14 mm button size on the smartphone had a 8.6% higher task completion rate and 40.7% lower time than the 8 mm button size when older adults used the screen mirroring with one-handed posture.

Acknowledgments

This work was supported by funding from the Natural Science Foundation of Chongqing (cstc2016jcyjA0406) and the National Natural Science Foundation of China (Grants no. 71661167006).

Disclosure of potential conflict of interest

The authors declare that there is no conflict of interest regarding the publication of this article.

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